1. Lab 05: Transport-Layer Services & UDP

Lab Name: Lab 05: Transport-Layer Services & UDP **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

The User Datagram Protocol (UDP) is a core Transport Layer protocol that provides a connectionless, lightweight communication service. Unlike TCP, UDP does not guarantee delivery, order, or error checking, making it highly efficient and suitable for time-sensitive applications like video streaming and DNS where occasional packet loss is acceptable. In this lab, you will generate UDP traffic within your CML topology and analyze its behavior, focusing on its low-overhead characteristics and understanding scenarios where connectionless transport is preferred.

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3. Objective / Purpose

To analyze Connectionless Transport (UDP) behavior, its lack of reliability mechanisms, and how it handles datagrams.

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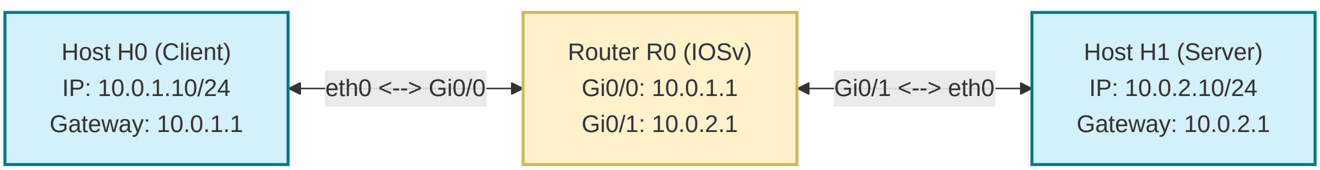
4. Topology Diagram



Details:

- 2x Hosts (`<initials>-H0` and `<initials>-H1`) communicating over a routed CML topology via Router (`<initials>-R0`).

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 2x Linux Nodes (configured as hosts), 1x IOSv Router

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6. Configuration / Methodology

IP Addressing Table

Device	CML Node Name	Interface	IP Address	Subnet Mask
Client	`<initials>-H0`	eth0	10.0.1.10	255.255.255.0
Server	`<initials>-H1`	eth0	10.0.2.10	255.255.255.0

Router	`<initials>-R0`	G0/0	10.0.1.1	255.255.255.0
Router	`<initials>-R0`	G0/1	10.0.2.1	255.255.255.0

Cisco IOSv Router Configuration Guide

This is your first time configuring a Cisco router using the Internetwork Operating System (IOS) command-line interface. Cisco IOS relies on hierarchical command modes. Understanding these modes is crucial for configuration:

1. **User EXEC Mode** (`Router>`): Default upon connecting. Limited to basic monitoring commands.
2. **Privileged EXEC Mode** (`Router#`): Provides detailed monitoring and administration commands. Entered by typing `enable` .
3. **Global Configuration Mode** (`Router(config)#`): Used to configure system-wide parameters (e.g., hostnames). Entered by typing `configure terminal` .
4. **Interface Configuration Mode** (`Router(config-if)#`): Used to configure specific physical/virtual ports. Entered from global configuration by typing `interface <interface-name>` (e.g., `interface GigabitEthernet0/0`).

Step-by-Step Router Setup:

Open the **Console** tab of your Router node in the CML interface, press **Enter** to initialize the command line, and configure the interfaces as follows:

1. **Elevate to Privileged EXEC Mode**:

   ```
   Router> enable
   ```

2. **Enter Global Configuration Mode**:

   ```
   Router# configure terminal
   ```

3. **Set the Hostname**: Change the default hostname to use your capitalized first and last initials in place of `KH` (do not type the `< >` brackets; for example, John Doe would type `hostname JD-R0`):

   ```
   Router(config)# hostname <initials>-R0
   ```

4. **Configure GigabitEthernet0/0 (facing Host H0)**:

   ```
   KH-R0(config)# interface GigabitEthernet0/0
   KH-R0(config-if)# ip address 10.0.1.1 255.255.255.0
   KH-R0(config-if)# no shutdown
   KH-R0(config-if)# exit
   ```

   (The `no shutdown` command turns on the virtual physical interface).

5. **Configure GigabitEthernet0/1 (facing Host H1)**:

   ```
   KH-R0(config)# interface GigabitEthernet0/1
   KH-R0(config-if)# ip address 10.0.2.1 255.255.255.0
   ```

```
KH-R0(config-if)# no shutdown
KH-R0(config-if)# exit
```

6. **Save Your Configurations**: Exit back to Privileged EXEC mode and copy the running state to startup memory:

   ```
   KH-R0(config)# end
   KH-R0# write memory
   ```

7. **Verify Router Interfaces and Routing Table**: Run the following verification commands to ensure the interfaces are active and the routes are automatically generated:

   - `show ip interface brief` (Check that both `Gi0/0` and `Gi0/1` show status/protocol as **up/up**).
   - `show ip route` (Ensure you see two connected networks (`C`) for `10.0.1.0/24` and `10.0.2.0/24` in the routing table).

   > ***IMPORTANT - Deliverable Screenshot****: Take a screenshot of your router CLI console displaying the output of both the `show ip interface brief` and `show ip route` commands. You must include this screenshot in your final lab report.*

Traffic Generation & Capture Instructions

1. **Start the Listener on Host H1**: Open H1's console and start netcat in UDP listen mode on port `5000` :

   ```
   nc -ul -p 5000
   ```

2. **Start Packet Capture in CML**: Right-click the link between **Host H0 and Router R0**, or between **Router R0 and Host H1**, select **Packet Capture**, and click **Start**.
3. **Send UDP Packets from Host H0**: Open H0's console and use netcat to send messages to H1:

   ```
   nc -u 10.0.2.10 5000
   ```

   Type a few text messages and press **Enter** (you should see them output on H1's console). When done, press `Ctrl+C` to terminate netcat.
4. **Download the Capture**: Stop the packet capture in CML and download the PCAP for analysis.

Narrative

UDP does not establish a connection before sending data. By using a tool like netcat (`nc -u`) to send messages, we can observe UDP datagrams in Wireshark without any handshake.

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7. Wireshark Analysis and Questions

Open your downloaded packet capture in Wireshark, locate the UDP traffic generated by your netcat session, and answer the following questions:

1. Select the first UDP segment sent from Host H0 to Host H1 in your packet capture.
   - What is the Wireshark packet number of this segment?
   - What type of application-layer payload or protocol message is being carried in this UDP segment?
   - Expand the User Datagram Protocol header in the packet details pane. How many fields are there in the UDP header, and what are their names?

2. By consulting the packet details pane in Wireshark (or the textbook), what is the length (in bytes) of each of the four UDP header fields?
3. The value in the UDP `Length` field is the length of what? Verify this claim with your captured UDP packet by calculating the total bytes of the UDP header plus the payload, and comparing it to the value shown in the Length field.
4. What is the maximum number of bytes that can be included in a UDP payload? (Hint: An IPv4 datagram has a maximum total length of 65,535 bytes. Consider the minimum size of an IPv4 header and the size of the UDP header).
5. What is the largest possible port number for a UDP source or destination port? (Hint: Consider the bit-width allocated to the port fields).
6. What is the protocol number for UDP? Give your answer in decimal notation. (Hint: To find this, inspect the `Protocol` field in the IPv4 header of the IP datagram carrying your UDP segment).
7. Locate a pair of UDP packets in your capture where you sent a message from Host H0 to Host H1, and then typed a response on Host H1's console to send back to Host H0.
    - What is the packet number of the first UDP segment (H0 -> H1)? What are its source and destination port numbers?
    - What is the packet number of the second UDP segment (H1 -> H0)? What are its source and destination port numbers?
    - Describe the mathematical/logical relationship between the port numbers in these two packets.

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Confirmed the lightweight, connectionless nature of UDP, which is ideal for time-sensitive applications like video streaming where slight loss is acceptable but delay is not.

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10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
    - Your written answers to the 7 Wireshark analysis questions in Section 7.
    - The required deliverable screenshot of the router CLI console showing the interface and routing tables as specified in Step 7 of the configuration guide.
    - A screenshot of your active CML topology showing the running hosts (`<initials>-H0` , `<initials>-H1`) and router (`<initials>-R0`).
    - An annotated screenshot of your Wireshark capture window showing the first captured UDP packet.